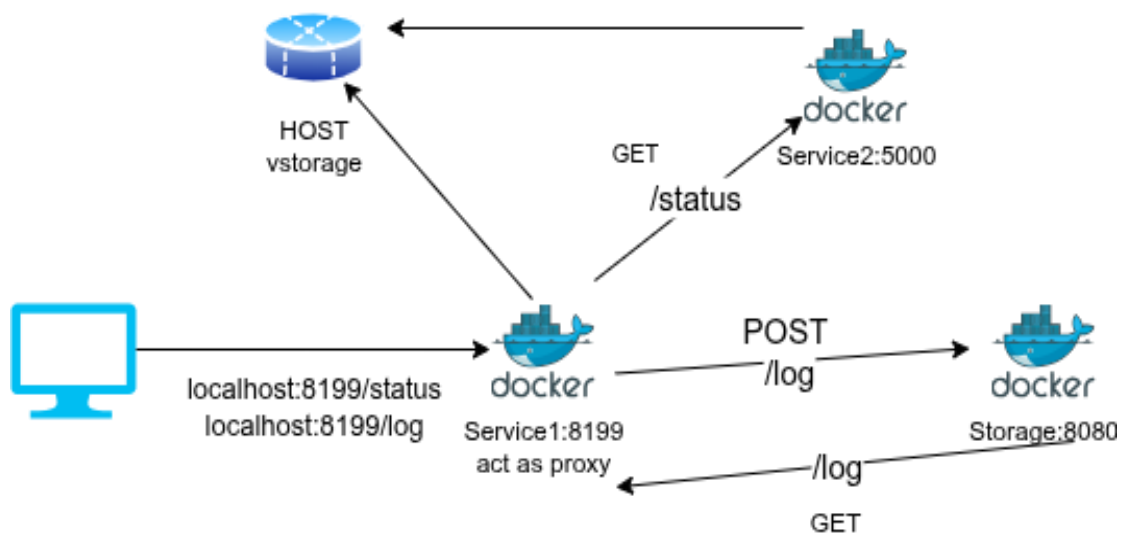


Basic information about the platform used:

Host machine: Debian 12, running on core i5, 16GiB Ram

Docker version 28.4.0, compose version 2.39.4

System Diagram:

IP address is omitted because I used bridged network, and it change on each launch. In the application, I call each service using service name specified in docker-compose

Instructions:

- git clone -b exercise1 https://github.com/iasiff/Tuni_DevOps.git
- After cloning, please open the terminal in the **Tuni_DevOps** directory and run **mkdir vstorage**
- **docker-compose up**
- **curl localhost:8199/status**
- **cat vstorage/log.txt**

You will see the desired output.

Analysis of the content of the status records:

Which disc space and uptime actually measured?

While running the containers it showed the uptime of underlying **host machine uptime** and **host disk space**. Docker shares the host kernel information.

How relevant are those measurements? What should be done better?

The measurement is important for 'running container', its underlying disk space and uptime is sometimes necessary. It would be better to show the actual container's uptime and disk space. However, this requires changes in code. Docker with grafana monitoring can be used to measure individual metrics of host and containers.

Analysis and comparison of the persistent storage solutions – what is good and what is bad in each solution?

It would be better to use vStorage rather than only in application local storage. The vstorage is good because it keeps the logs. However, for local storage, if the application turns off the logs might gone.

Teacher's instruction for cleaning up the persistent storage:

- **docker compose down**
- **echo y | docker system prune**

To remove vstorage from Teacher's host machine:

```
sudo rm -rf vstorage/
```

What was difficult?

The difficult part was coding in another programming language and framework which I have not done before and searching for different functions in that language. Also writing Dockerfile for that programming language with best practice possible

What were the main problems?

No serious problem faced while doing the exercise. Just needed to understand the system takes time.

Attachment in repository:

llm.txt

docker-status.txt

